FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 186**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

1. ***Section-A (01 – 08, 19 – 26, 37 – 44)*** *contains 24 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Full Marks : **+4** If only (all) the four option(s) is (are) chosen.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen and both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–1** In all other cases.*

   Section-A (09 – 12, 27 – 30, 45 – 48) *contains 6 List-Match sets with 12 questions (each set has 2 questions). Each question has 4 statements in LIST-I & 6 statements in LIST-II. The codes for Lists have choices (A), (B), (C), (D) out of which only one is correct.*
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-C (13 – 18, 31 – 36, 49 – 54)*** *contains 18 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

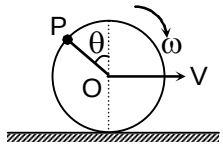
PART – I

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A cubical block of wood of edge 10cm and mass 0.92 kg floats on a tank of water with oil of rel. density 0.6 to a depth of 4 cm above water. When the block attains equilibrium with four of its sides edges vertical
(A) 1 cm of it will be above the free surface of oil
(B) 5 cm of it will be under water
(C) 2 cm of it will be above the common surface of oil and water
(D) 8 cm of it will be under water

2. A disc of radius *R* rolls on a horizontal surface with linear velocity *V* and angular velocity ω. There is a point *P* on circumference of disc at angle θ with upward vertical diameter measured anticlockwise see figure, which has a vertical velocity. Here θ is equal to



(A) $\pi + \sin^{-1}\frac{V}{R\omega}$ (B) $\frac{\pi}{2} - \sin^{-1}\frac{V}{R\omega}$

(C) $\pi - \cos^{-1}\frac{V}{R\omega}$ (D) $\pi + \cos^{-1}\frac{V}{R\omega}$

3. A point moves with deceleration along the circle of radius R so that at any time its tangential and normal acceleration are equal in magnitude. At the initial moment t =0, the velocity of the point is v_0. The velocity of the point will be

(A) $v = \frac{v_0}{1+\frac{v_0 t}{R}}$ at t second (B) $v = v_0 e^{-S/R}$ after displacements

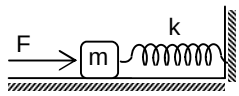
(C) $v = v_0 e^{-SR}$ after displacements (D) $v = \frac{v_0}{1-\frac{v_0 t}{R}}$ at t second

Space for Rough work

4. A horizontal disc rotates freely about a vertical axis through its centre. A ring, having the same mass and radius as the disc, is now gently placed on the disc. After some time, they rotate with a common angular speed.
(A) some friction exists between the disc and the ring.
(B) the angular momentum of the disc plus ring is conserved.
(C) the final common angular speed is $\frac{2}{3}$ rd of the initial angular velocity of the disc.
(D) $\frac{2}{3}$ rd of the initial kinetic energy changes to heat.

5. If electron of the hydrogen atom is replaced by another particle of same charge but of double the mass, then:
(A) Bohr radius will increase
(B) ionisation energy of the atom will be doubled
(C) speed of the new particle in a given state will be lesser than the electron's speed in same orbit
(D) gap between energy levels will now be doubled

6. A constant force F is applied on a spring block system as shown in figure. The mass of the block is m and spring constant is k. The block is placed over a smooth surface. Initially the spring was unstretched. Choose the correct alternative(s).

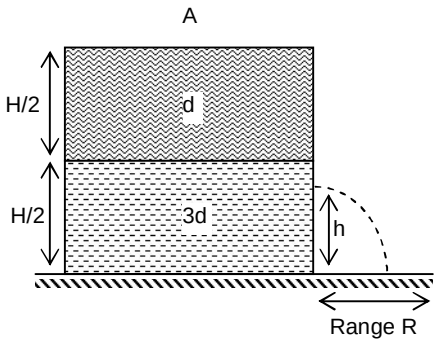


(A) the block will execute SHM
(B) amplitude of oscillation is F/2k
(C) time period of oscillation is $2\pi\sqrt{\frac{m}{k}}$
(D) the maximum speed of block is $\sqrt{\frac{2Fx - kx^2}{m}}$
(here x = F/k)

Space for Rough work

7. For an ideal gas:
(A) the change in internal energy in a constant pressure process from temperature T_1 to T_2 is equal to $nC_v (T_2 - T_1)$, where C_v is the molar specific heat at constant volume and n the number of moles of the gas.
(B) the change in internal energy of the gas and the work done by the gas are equal in magnitude in an adiabatic process.
(C) the internal energy does not change in an isothermal process.
(D) no heat is added or removed in an adiabatic process.

8. A container of large uniform cross sectional area A resting on a horizontal surface holds two immiscible non viscous and incompressible liquids of density d and 3d, each of height H/2. The lower density liquid is open to the atmosphere having pressure P_0. A tiny hole of area a(a<<A) is punched to the vertical side of lower container at a height h (0<h<H/2) for which range is maximum.



(A) h = H /3

(B) Range R = $\frac{2H}{3}$

(C) Range R = $\frac{3H}{2}$

(D) Velocity of efflux $v = \sqrt{\frac{2}{3}gH}$

Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

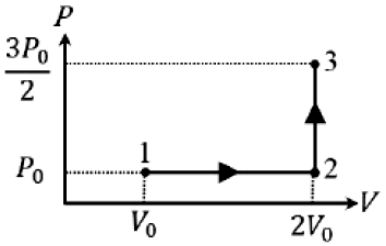
Answer the following by appropriately matching the list based on the information given in the paragraph.

In the thermodynamic process on an ideal monatomic gas, the infinitesimal heat absorbed by the gas is given by $T\Delta X$. where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas $X = \frac{3}{2}R\ln\left(\frac{T}{T_A}\right) + R\ln\left(\frac{V}{V_A}\right)$. Here R is gas constant. V is volume of gas. T_A and V_A are constants.

The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

List – I		List – II	
(I)	Work done by the system in process $1\rightarrow2\rightarrow3$	(P)	$\frac{1}{3}RT_0\ln 2$
(II)	Change in internal energy in process $1\rightarrow2\rightarrow3$	(Q)	$\frac{1}{3}RT_0$
(III)	Heat absorbed by the system in process $1\rightarrow2\rightarrow3$	(R)	RT_0
(IV)	Heat absorbed by the system in process $1\rightarrow2$	(S)	$\frac{3}{3}RT_0$
		(T)	$\frac{1}{3}RT_0(3+\ln 2)$
		(U)	$\frac{5}{6}RT_0$

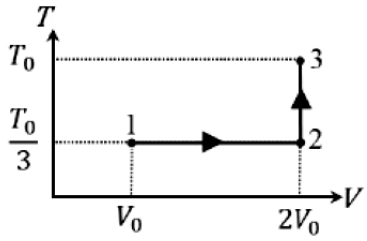
9. If the process carried out on one mole of monatomic ideal gas is as shown in figure in the PV-diagram with $P_0V_0 = \frac{1}{3}RT_0$, the correct match is.



(A) I $\rightarrow$ Q, II $\rightarrow$ R, II I$\rightarrow$ S, IV $\rightarrow$ U
(B) I $\rightarrow$ Q, II $\rightarrow$ S, III $\rightarrow$ R, IV $\rightarrow$ U
(C) I $\rightarrow$ Q, II $\rightarrow$ R, III $\rightarrow$ P, IV $\rightarrow$ U
(D) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ Q, IV $\rightarrow$ T

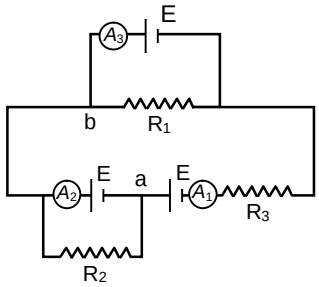
Space for Rough work

10. If the process carried out on one mole of monatomic ideal gas is as shown in figure in the PV-diagram with $P_0V_0 = \frac{1}{3}RT_0$, the correct match is.



(A) I $\rightarrow$ P, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ T
(B) I $\rightarrow$ S, II $\rightarrow$ T, III $\rightarrow$ Q, IV $\rightarrow$ U
(C) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ S
(D) I $\rightarrow$ P, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ P

Referring to the given circuit match List-I with List-II in question no. 11 and 12.



List – I		List – II	
(I)	Reading of ammeter A_1 in ampere is	(P)	4/3
(II)	Reading of ammeter A_2 in ampere is	(Q)	8/3
(III)	Reading of ammeter A_3 in ampere is	(R)	4
(IV)	Potential difference between point a and point b in volt is	(S)	zero
		(T)	2
		(U)	–1

11. In the above circuit, $R_1 = R_2 = R_3 = 3\Omega$ and e.m.f. of each cell is E = 4V and negligible internal resistance. All ammeters are ideal. Match the following:
(A) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ R, IV $\rightarrow$ S (B) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ Q
(C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ R (D) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ U

Space for Rough work

12. In the above circuit if R_2 is removed and $R_1 = R_3 = 4\ \Omega$

(A) I $\rightarrow$ P, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ R (B) I $\rightarrow$ T, II $\rightarrow$ P, III $\rightarrow$ S, IV $\rightarrow$ Q

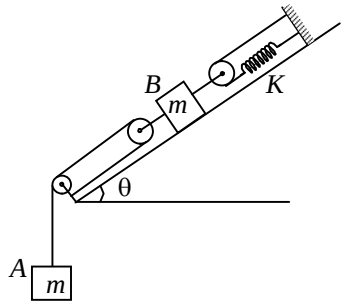
(C) I $\rightarrow$ P, II $\rightarrow$ Q, III $\rightarrow$ S, IV $\rightarrow$ R (D) I $\rightarrow$ S, II $\rightarrow$ R, III $\rightarrow$ T, IV $\rightarrow$ U

SECTION – C
(Numerical Answer Type)

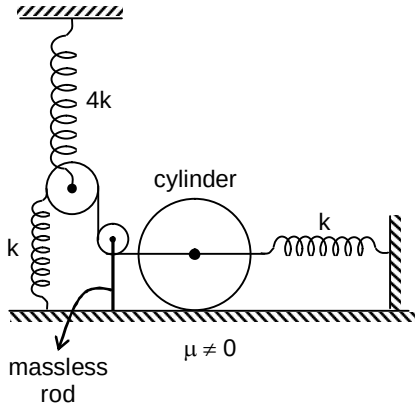
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

13. A uniform disc of radius R having charge Q distributed uniformly all over its surface is placed on a smooth horizontal surface. A magnetic field, $B = kxt^2$, where k is a constant, x is the distance (in metre) from the centre of the disc and t is the time (in second), is switched on perpendicular to the plane of the disc. Find the torque (in N-m) acting on the disc after 15 sec. (Take 4kQ = 2.50 S.I. unit and R = 1m).

14. Two blocks *A* and *B*, each of mass *m* are connected by means of a pulley-spring system on a smooth inclined plane of inclination θ as shown in the figure. All the pulleys and spring are ideal. Now, *B* is slightly displaced from its equilibrium position. It starts to oscillate. Time period of oscillation of *B* will be (Take *m* = 4 kg, *K* = 5 N/m, $\pi = 3.14$)

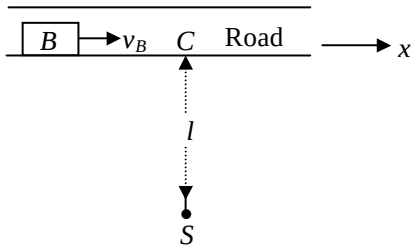


15. A cylinder of mass m = 3 kg and radius R = 2 m is attached to a massless spring-pulley system as shown in the figure and all the springs are in their natural length at t = 0. The friction is sufficient to cause pure rolling of the cylinder. What is the time period (in sec.) of cylinder for small horizontal displacement? (spring constant k = 2 N/m and $\pi = 3.14$)

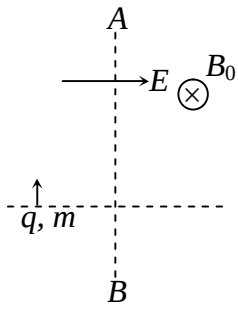


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16. A bus B is moving with a velocity v_B in the positive x-direction along a road as shown in the figure. A shooter 'S ' is at a distance l from the road. He has a detector which can detect signals only of frequency 1500Hz. The bus blows horn of frequency 1000 Hz. When the detector detects a signal the shooter immediately shoots towards the road along SC and the bullet hits the bus. Find the velocity of the bullet if velocity of sound in air is v = 340 m/s and $\frac{v_B}{v}=\frac{2}{3\sqrt{3}}$.



17. A uniform electric field *E* is present horizontally along the paper throughout region but uniform magnetic field B_0 is present horizontally (perpendicular to plane of paper in inward direction) right to the line *AB* as shown. A charge particle having charge *q* and mass *m* is projected vertically upward and crosses the line *AB* after time t_0. The speed of projection if particle moves after t_0 with constant velocity. (given *qE* = *mg*) is nt_o. Find the value of 'n'. ($g = 9.8\ m/s^2$)



18. In a moving coil galvanometer, torque on the coil can be expressed as $\tau \propto i$, where i is current through the wire. The rectangular coil of the galvanometer having number of turns N, area A and moment of inertia I is placed in magnetic field B. The torsion constant of spring in the galvanometer is C and charge Q is passed through the coil almost instantaneously. If the maximum angle through which coil deflected is $\frac{KNABQ}{\sqrt{4CI}}$. (Ignore the damping in mechanical oscillations).

Space for Rough work

Chemistry

PART – II

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. The bond order of which of the following substance(s) is/are greater than one?
(A) BF_3 (B) NH_3
(C) CO_2 (D) PCl_5

20. In which of the following reaction(s), carbocation intermediate(s) is/are formed
(A) $CH_3 - CH = CH_2 + HBr \longrightarrow$ (B) $CH_3 - CH = CH_2 + HBr \xrightarrow{(CH_3CO)_2O_2}$
(C) $CH_3 - CH = CH_2 + NBS \xrightarrow{\text{Radiation}}$ (D) $CH_3 - CH = CH_2 \xrightarrow[NaBH_4]{Hg(OCOCH_3)_2,\ H_2O}$

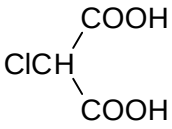
21. Which of the following nucleophile(s) can form products that show functional isomerism?
(A) CN^- (B) SH^-
(C) NO_2^- (D) NH_3

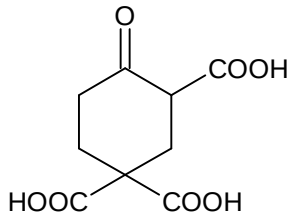
22. $Zn(s) \mid Zn^{2+}(C_1) \parallel Ag^+(C_2) \mid Ag(s)$
Under which of the following conditions, E_{Cell} will be equal to E^0_{Cell} ?
(A) $C_1 = C_2 = 1$ M (B) $C_2 = \sqrt{C_1}$
(C) $C_1 = C_2 = 0.1$ M (D) $C_2 - C_1 = 1$ M

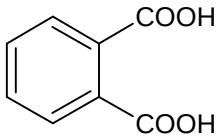
23. The correct statement(s) regarding the halides of boron is/are
(A) BI_3 undergo the easiest hydrolysis than the other halides
(B) $BF_3 + BCl_3 \longrightarrow BF_2Cl + BFCl_2$
(C) Inorganic benzene is formed when BCl_3 reacts with NH_4Cl followed by reaction with $NaBH_4$
(D) Lewis acidity order: $BBr_3 > BCl_3 > BF_3$

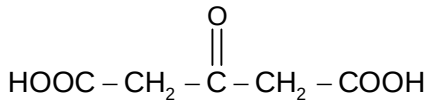
Space for Rough work

24. Which of the following substituted polycarboxylic acids are converted to their corresponding monocarboxylic acids on heating?

(A) 

(B) 

(C) 

(D) 

25. Which of the following salts produce(s) gas(es) when treated with dilute mineral acids?
(A) $BaCO_3$ (B) K_2SO_3
(C) $Ca_3(PO_4)_2$ (D) KNO_2

26. Which of the following reaction(s) is/are exothermic in nature?
(A) $CH_2 = CH_2(g) + H_2(g) \xrightarrow{Ni} C_2H_6(s)$ (B) $NaOH + HCN \longrightarrow NaCN + H_2O$
(C) $HCl + H_2O \longrightarrow H^+(aq) + Cl^-(aq)$ (D) $H_2(g) \longrightarrow H(g) + H(g)$

Space for Rough work

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

List-I contains ionic solids and list-II states regarding their crystal structure.

	List-I		List-II
(I)	NaCl	(P)	Cations occupy half of the tetrahedral voids of FCC unit cell
(II)	CsCl	(Q)	Cations occupy octahedral voids of FCC unit cells
(III)	ZnS	(R)	Coordination number of the cation is eight
(IV)	CaF_2	(S)	Cations and anions have different coordination numbers
		(T)	Coordination number of both cations and anions is four
		(U)	Anions occupy the voids of the unit cell

27. The correct combination out of the following is

(A) (I) $\rightarrow$ (T) (B) (III) $\rightarrow$ (Q)

(C) (II) $\rightarrow$ (R) (D) (IV) $\rightarrow$ (P)

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(IV)	CaF_2	(S)	Cations and anions have different coordination numbers
		(T)	Coordination number of both cations and anions is four
		(U)	Anions occupy the voids of the unit cell

28. Which of the following combination is correct

(A) (IV) $\rightarrow$ (Q) (B) (III) $\rightarrow$ (P)

(C) (I) $\rightarrow$ (U) (D) (II) $\rightarrow$ (S)

Space for Rough work

List-I contains complexes and their behaviour is mentioned in list-II.

	List-I		List-II
(I)	$[Cu(NH_3)_4(H_2O)_2]^{2+}$	(P)	Colourless complex
(II)	$[FeF_6]^{3-}$	(Q)	Back donation from metal to ligand takes place
(III)	$[Cr(en)_3]^{3+}$	(R)	Prone to Jahn Teller distortion
(IV)	$[Ni(CO)_4]$	(S)	Square planar complex
		(T)	Chelate complex
		(U)	Coordination number is six

29. The correct combination out of the following is

(A) (I) $\rightarrow$ (T) (B) (II) $\rightarrow$ (P)

(C) (III) $\rightarrow$ (Q) (D) (IV) $\rightarrow$ (S)

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(IV)	$[Ni(CO)_4]$	(S)	Square planar complex
		(T)	Chelate complex
		(U)	Coordination number is six

30. The correct combination out of the following is

(A) (I) $\rightarrow$ (R) (B) (II) $\rightarrow$ (Q)

(C) (III) $\rightarrow$ (P) (D) (IV) $\rightarrow$ (T)

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

31. What volume in mL of 0.5 M acidified potassium permanganate solution can completely oxidize 523.15×10^{-3} moles of Sn^{2+} ions to Sn^{4+} ions?
$MnO_4^- + Sn^{2+} + H^+ \longrightarrow Mn^{2+} + Sn^{4+} + H_2O$

32. The kinetic energy of a moving electron is 50 kJ. If the de-Broglie wavelength associated with the electron motion is expressed as $y \times 10^{-z}$ m, and the value of z is 20. What is the value of (y + z)?
[Assume that $m_e = 9 \times 10^{-31}$ kg and $h = 6.6 \times 10^{-34}$ Js]

33. Reaction of phenol(PhOH) with bromine water(Br_2/H_2O) forms a precipitate. What is the molar mass of the precipitate in g mol^{-1} unit?
[Molar mass of bromine = 79.9]

34. One mole of ethylene glycol was dissolved in 640.8 g water and the resulting solution was cooled to -3.72°C temperature. How many gram of ice is formed due to freezing of water?
[K_f = 1.86 K kg mol^{-1}]

35. Treatment of NaCl with acidified solution of (X) forms a red vapour(Y). Treatment of (Y) with aqueous NaOH forms a yellow solution(Z). $Pb(CH_3COO)_2$ gives a yellow precipitate when treated with solution of (Z). What is the molar mass of (Y) in g mol^{-1} unit? [At. Mass of Cr = 51.9]

36. How much current in ampere unit is to be passed through $CuSO_4$ for four minutes in order to deposit 190.5 g copper on the unattacked cathode?

Space for Rough work

Mathematics

PART – III

SECTION – A
(One or More than one correct type)

This section contains **08** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

37. If the pair of asymptotes of a hyperbola H is $xy - 2x - y + 2 = 0$. Its conjugate hyperbola H' passes through (3, 0) then

(A) eccentricity of H' is $\sqrt{2}$

(B) focus of H' is $\left(2+\frac{1}{\sqrt{2}}, 2+\frac{1}{\sqrt{2}}\right)$

(C) focus of H is $\left(\sqrt{2}, -\sqrt{2}\right)$

(D) equation of transverse axis of H is $x - y + 1 = 0$

38. If $\sum_{r=0}^{n} \frac{2r}{{}^{n}C_{r}} = \sum_{r=0}^{n} \frac{n^2 - 3n + 3}{{}^{n}C_{r}}$, then

(A) n = 1 (B) n = 2

(C) n = 3 (D) n = 4

39. The values of m for which it is possible to draw the chord $y = \sqrt{m}x + 1$ to the curve $x^2 + 2xy + \left(2 + \sin^2\alpha\right)y^2 = 1$, which subtends of right angle at origin for some value of α is/are

(A) m = 2 (B) $m = \frac{3}{2}$

(C) $m = 3$ (D) $m = \frac{7}{2}$

Space for Rough work

40. If y = f(x) and y = g(x) are symmetrical about the line x = $\frac{\alpha+\beta}{2}$, then $\int_{\alpha}^{\beta} f(x)\, g'(x)\, dx$ is equal to

(A) $\int_{\alpha}^{\beta} f'(x)\, g(x)\, dx$ (B) $-\int_{\alpha}^{\beta} f'(x)\, g(x)\, dx$

(C) $\frac{1}{2}\int_{\alpha}^{\beta} (f(x)\, g'(x) - f'(x)\, g(x))\, dx$ (D) $\frac{1}{2}\int_{\alpha}^{\beta} (f(x)\, g'(x) + f'(x)\, g(x))\, dx$

41. If the probability of choosing an integer 'k' out of 2m integers 1, 2, 3,……..2m is inversely proportional to $k^4\,(1 \le k \le m)$. If x_1 is the probability that choosen number is odd and x_2 is the probability that chosen number is even, then

(A) $x_1 < \frac{2}{3}$ (B) $x_1 > \frac{1}{2}$

(C) $x_1 > \frac{2}{3}$ (D) $x_2 < \frac{1}{2}$

42. Which of the following functions are twice differentiable at x = 0?

(A) $f(x) = x\,|x|$ (B) $g(x) = [x^2]\tan^{-1}x - \{x^2\}\cot^{-1}x - [x^2]\frac{\pi}{2}$

(C) $h(x) = |\sin^2 x|$ (D) $h(x) = \begin{cases} x^4 \cos\left(\frac{1}{x}\right); & x \neq 0 \\ 0 \quad\quad\quad\quad ; & x = 0 \end{cases}$

43. Let z_1, z_2, z_3 be the coordinate of vertices A, B, C of the triangle if $W_1 = z_1 - z_2$, $W_2 = z_3 - z_2$ and $\angle A = 90°$, then

(A) $\mathrm{Re}\left(\frac{W_2}{W_1}\right) = 1$ (B) $\mathrm{Re}(W_1\bar{W}_2) = |W_1|^2$

(C) $\mathrm{Re}(W_1\bar{W}_2) = |W_2|^2$ (D) $\mathrm{Re}\left(\frac{W_1}{W_2}\right) = 1$

Space for Rough work

44. If $A(\alpha,\beta)=\begin{bmatrix}\cos\alpha & \sin\alpha & 0\\ -\sin\alpha & \cos\alpha & 0\\ 0 & 0 & e^{\beta}\end{bmatrix}$, then

(A) $A(\alpha,\beta)' = A(-\alpha, \beta)$ (B) $A(\alpha, \beta)^{-1} = A(-\alpha,-\beta)$

(C) $\text{Adj}(A(\alpha,\beta)) = e^{\beta}A(-\alpha, -\beta)$ (D) $A(\alpha,\beta)' = A(\alpha,-\beta)$

(Matching List Type)

This section contains **TWO (02)** List-Match Sets. Each List-Match set has **TWO(02)** Multiple Choice Questions. Each List-Match set has two lists: **List-I** and **List-II**. **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U). **FOUR** options are given in each Multiple Choice Question based On **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.

Answer the following by appropriately matching the list based on the information given in the paragraph.

A plane π passes through the point (1, 1, 1) and is parallel to the vectors $\vec{b}=(1,0,-1)$ and $\vec{a}=(-1,1,0)\pi$ meets the axes in A, B and C forming the tetrahedron OABC.

	List-I		List-II
(I)	Volume of the tetrahedron OABC in cubic units.	(P)	$\frac{9}{2}$
(II)	Area of a face of the tetrahedron in square units.	(Q)	$\frac{9\sqrt{3}}{2}$
(III)	Distance of the points, $\left(\frac{3\sqrt{3}}{2}, 3\sqrt{3}, 3\right)$ from the plane π in units.	(R)	$\sqrt{3}$
(IV)	Distance of a face of the tetrahedron from the vertex	(S)	3
		(T)	4
		(U)	$\frac{1}{2}$

45. Which of the following is the only correct combination?

(A) II$\rightarrow$U (B) II$\rightarrow$T

(C) I$\rightarrow$P (D) I$\rightarrow$S

Space for Rough work

46. Which of the following is the only correct combination?
(A) IV→Q (B) III→R
(C) IV→S (D) III→Q

Answer the following by appropriately matching the list based on the information given in the paragraph.

Suppose a, b, c are three distinct real numbers and f(x) is a real quadratic polynomial such that

$$\begin{bmatrix} 4a^2 & 4a & 1 \\ 4b^2 & 4b & 1 \\ 4c^2 & 4c & 1 \end{bmatrix}\begin{bmatrix} f(-1) \\ f(1) \\ f(1) \end{bmatrix} = \begin{bmatrix} 3a^2+3a \\ 3b^2+3b \\ 3c^2+3c \end{bmatrix}$$

List-I		List-II	
(I)	x – coordinates (s) of the point of intersection of $y = f(x)$ with the x – axis	(P)	4
(II)	Area bounded by $y = \frac{3}{2}f(x)$ and the x – axis	(Q)	2
(III)	Maximum value of $f(x)$	(R)	1
(IV)	Length of the intercept made by $y = f(x)$ on the x – axis	(S)	–2
		(T)	0
		(U)	3

47. Which of the following is the only correct combination?
(A) II→U (B) II→P
(C) I→P (D) I→U

48. Which of the following is the only correct combination?
(A) IV→Q (B) III→R
(C) IV→S (D) III→Q

Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. XXXXX.XX).

49. Four students appeared in three examinations. The number of possible ways in which we can have result, such that neither any students fails in all the examinations nor there is an examination which is not cleared by any student is k, then find $\frac{k}{100}$

50. The straight line $\frac{x}{4}+\frac{y}{3}=1$ intersects the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ at two points *A* and *B*, there is a point *P* on this ellipse such that the area of ΔPAB is equal to $6(\sqrt{2}-1)$, the number of such points (*P*) is

51. Let $\vec{b}=4i+3j$ and $\vec{c}$ be two vectors perpendicular to each other in the xy – plane. If $\vec{r}_i, i=1,2,....n$ are the vectors in the same plane having projection 1 and 2 along b and c respectively $\sum_{i=1}^{n}\left|\vec{r}_i\right|^2$ is equal to k then $\frac{k+20}{3}$ equals to

52. If $I=\int_{1/e}^{e}|\log x|\frac{dx}{x^2}$, then $\frac{I}{4}+\frac{1}{2e}=$_____

53. If $f(x)+f(y)=f\left(\frac{x+y}{1-xy}\right)$ for all $x,y\in R$, $xy\neq 1$ and $\lim_{x\to 0}\frac{f(x)}{x}=2$, then the value of $\frac{f\left(\sqrt{3}\right)}{f'(-2)}$ is

54. The minimum value of $(\tan A-\cos B)^2+(\cot A-\sin B)^2$, where A and B are independent variables, is of the form $a-b\sqrt{c}$, where a, b, c are natural numbers. The value of $a+b+c$ is equal to

Space for Rough work